Beyond Schrödinger-Poisson: Nonrelativistic EFT for Scalar Dark Matter

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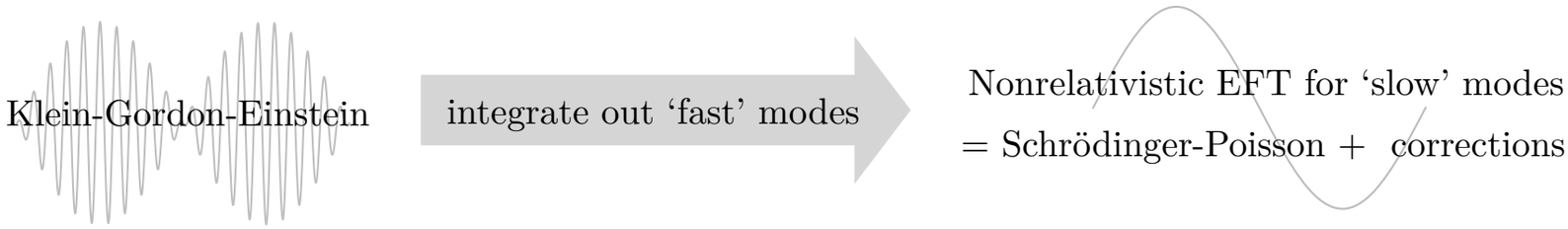
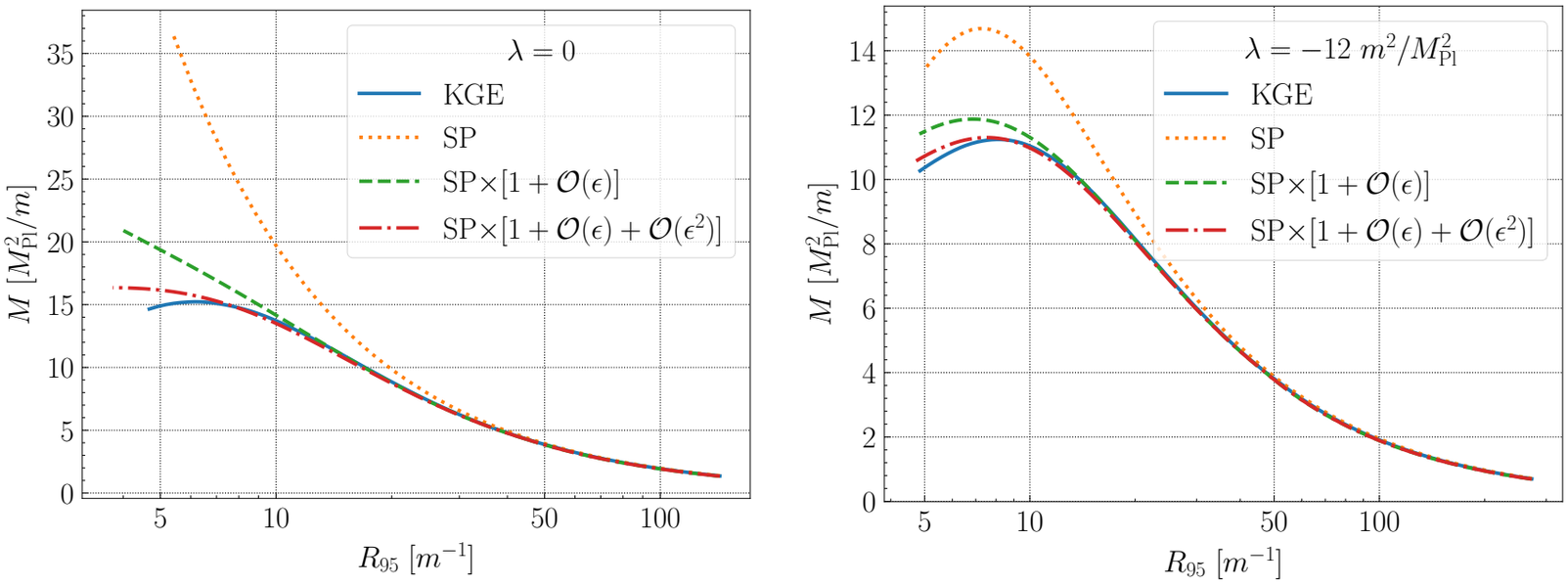


$$
\begin{aligned}
& i\dot{\psi}_s + \frac{3i}{2}H_s\psi_s + \frac{1}{2ma_s^2}\nabla^2\psi_s - m\Phi_s\psi_s - \frac{\lambda}{8m^2}|\psi_s|^2\psi_s \\
& + \left(\frac{3\rho_s}{8mM_{\rm Pl}^2} + \frac{|\bar{\psi}_s|^2}{2M_{\rm Pl}^2} + \frac{|\psi_s|^2}{16M_{\rm Pl}^2} - \frac{m}{2}\Phi_s^2\right)\psi_s - 2i\dot{\Phi}_s\psi_s + \frac{\nabla^4\psi_s}{8m^3a_s^4} \\
& + 3\Phi_s\frac{\nabla^2\psi_s}{2ma_s^2} - \frac{\nabla\Phi_s\cdot\nabla\psi_s}{2ma_s^2} - i\frac{\vec{\sigma}_s\cdot\nabla\psi_s}{a_s} + \left(\frac{17\lambda^2}{8m^2} - \frac{\kappa}{\Lambda^2}\right)\frac{|\psi_s|^4\psi_s}{96m^3} \\
& - \frac{\lambda}{16m^4a_s^2}\left(2|\nabla\psi_s|^2\psi_s + \psi_s^2\nabla^2\psi_s^* + 2|\psi_s|^2\nabla^2\psi_s + \psi_s^*(\nabla\psi_s)^2\right) = 0 + \mathcal{O}(\epsilon^4)
\end{aligned}
$$

nonlinear Schrödinger eq.

+ corrections

KGE = Klein-Gordon-Einstein, SP = Schrödinger-Poisson. Along with SP + corrections, we also have Poisson + corrections, as well as equations for other metric perturbations

soliton mass-radius relationship — an application of the EFT